

Fan Yang

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B.Eng. Student in Automation, Shanghai Jiao Tong University



Education

Shanghai Jiao Tong University | Automation | B.Eng. Student

2023.09 - Present

- GPA: 3.95/4.3; Rank: 12/99; Average: 91.14/100
- English Proficiency: IELTS 7.5; CET-6 643
- Relevant Coursework: Calculus (94), Linear Algebra (95), Linear & Nonlinear Programming (99), Programming (96), Data Structures (95), Automatic Control (99), Robotics (100)

Research Interests

Agentic Robotics • Multi-Agent / Multi-Robot Systems • Reinforcement Learning

Research Experience

Closed-Loop Traffic Flow Learning and Action-Aware Replanning for Multi-Robot Path Planning 2026.01 - 2026.06

- Problem: Studied congestion awareness and distribution shift in large-scale multi-robot path planning, using traffic-flow prediction for proactive replanning while mitigating model mismatch of offline-trained predictors in closed-loop deployment.
- Method: Proposed Action-Aware Refine & Replanning to explicitly model how candidate planning actions affect future traffic flow; combined with Loop Engineering to update traversal-time estimates and traffic-state models online using execution feedback.

Robust Multi-Robot Path Planning under Execution Errors 2025.06 - 2026.01

- Investigated path planning for decentralized multi-robot systems with execution errors; surveyed related work and designed a robust planner to improve feasibility and stability under uncertainty.
- Project: <https://github.com/fyang7705-sys/learn-to-follow-robust>

Publications

- Closed-Loop Traffic Flow Learning and Action-Aware Replanning for Multi-Robot Path Planning, IEEE Transactions on Systems, Man, and Cybernetics: Systems (TSMC), first-author manuscript submitted.

Projects

Semantic Guidance Robot for Visually Impaired Users in Public Indoor Spaces | Semantic Navigation Agent

2026.03 - Present

- Built a robotic framework for semantic guidance in airports, hospitals, and subway stations: natural language/image → semantic goal localization → autonomous navigation.
- Built the ROS framework, initial-pose estimation, and path planning; integrated keyframe semantic memory, a multi-agent collaboration framework, and ROS2 / Nav2 for end-to-end deployment and closed-loop real-robot validation on a low-cost platform (<https://intel-caragent.github.io/>).

RoboCup Soccer Robot SRC Club | Development & Maintenance

2023.12 - 2025.07

- Worked on modular organization and collaborative development of the full project; proficient in Git and CMake; built the open-source wiki for the RoboCup Small Size League team (<https://sjtu-src.github.io/Wiki/>).
- Implemented online calibration of robot kicking power using iterative least squares, improving execution-parameter estimation accuracy and control stability.

National Undergraduate Electronics Design Contest | Simple Self-Aiming Device

2025.07 - 2025.08

- Designed the mobile platform and aiming mechanism; used clustering algorithms and model training for target recognition, improving target-tracking capability in dynamic environments.

Ascend AI Innovation Competition | Image Detection System Development

2025.10 - 2025.12

- Surveyed image-detection methods and proposed a network architecture combining local-detail extraction with global semantic modeling; introduced an LLM to explain and analyze detection results, improving system interpretability.

Honors & Awards

- 2026 Mathematical Contest in Modeling (MCM), Honorable Mention
- 2025 Xiaomi Scholarship; SJTU Class B Scholarship
- 2025 National Undergraduate Electronics Design Contest, Provincial Second Prize
- 2025 China Robot Competition and RoboCup China Open, Second Prize
- 2024 China Robot Competition and RoboCup China Open, Runner-up
- 2024 SJTU Merit Student
- 2024 SJTU Class C Scholarship

Research Plan

During graduate study, I plan to investigate intelligent multi-robot decision-making in open, dynamic environments along the core pipeline of “environment prediction – collaborative planning – robust execution,” and develop a networked closed loop of “communication – prediction – decision – execution – feedback.” Near-term interests include efficient collaboration under bandwidth, latency, and packet-loss constraints, non-stationarity from policy coupling, and robustness to communication outages, node failures, and sudden task changes. Long term, I aim to integrate learning-based planning with classical optimization, control, and risk constraints to deploy real robot teams in logistics, intelligent transportation, and complex service environments.